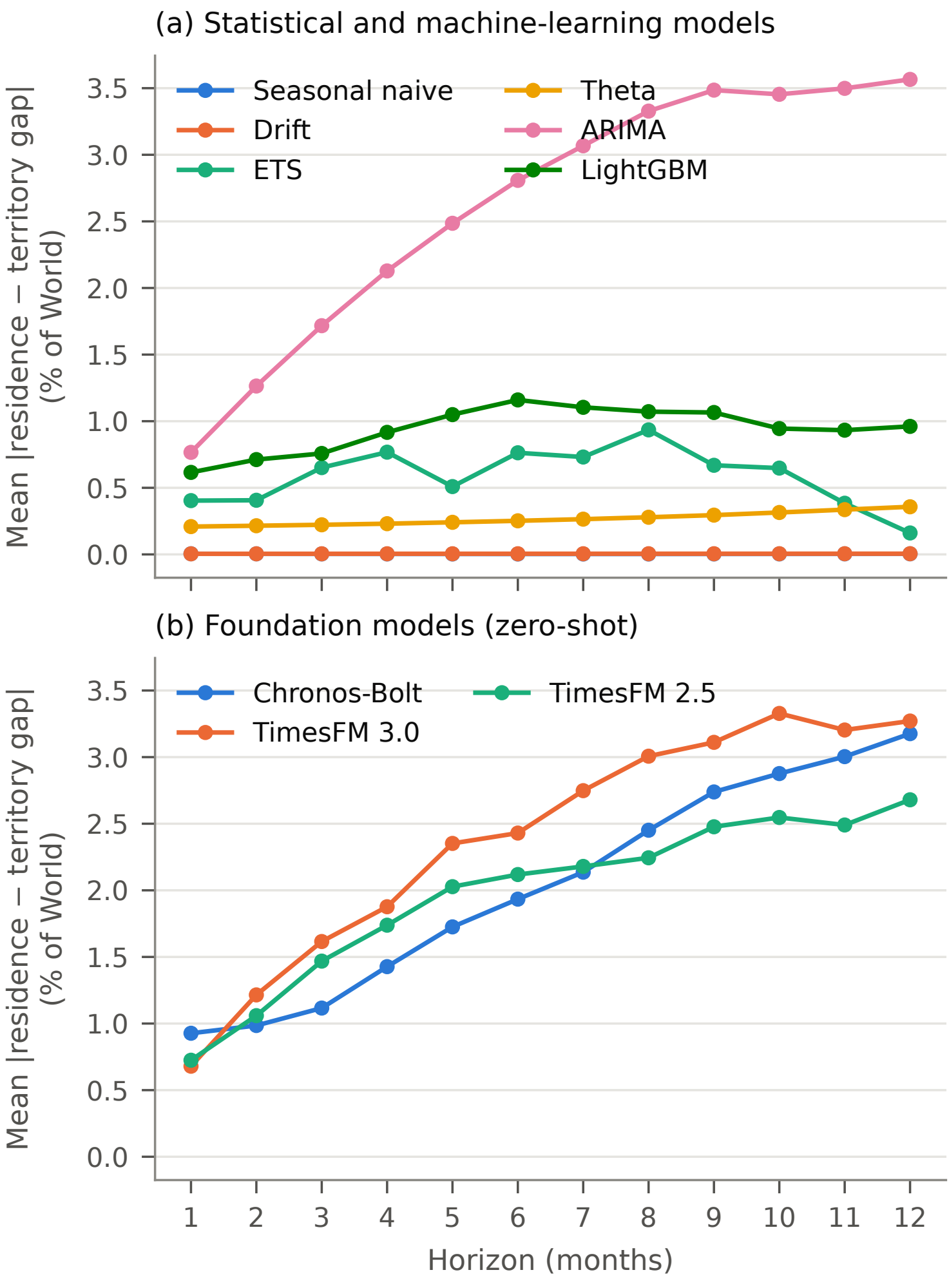
(a) Statistical and machine-learning models
Seasonal naive
Drift
ETS
Theta
ARIMA
LightGBM
Mean |residence − territory gap|
(% of World)
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
(b) Foundation models (zero-shot)
Chronos-Bolt
TimesFM 3.0
TimesFM 2.5
Mean |residence − territory gap|
(% of World)
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
1
2
3
4
5
6
7
8
9
10
11
12
Horizon (months)